Pranav Gupta

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SUMMARY

Aspiring AI Engineer with demonstrated experience building and deploying full-stack, LLM-powered applications from concept to production. Proven ability to lead hands-on engineering projects, including founding and managing a nonprofit solar car team and co-authoring a published ML research paper on autonomous driving. I enjoy tackling open-ended technical challenges and have spent 6+ years developing a wide range of Python projects and mentoring peers through leadership roles in solar car, robotics, rocketry, and math competitions.

EDUCATION

University of Pennsylvania, Philadelphia, Pennsylvania	Major: Artificial Intelligence BSE
Graduating in May 2029	Minor: Engineering Entrepreneurship
School of Science and Engineering (SEM), Dallas, Texas	Weighted GPA: 4.58
Graduated in May 2025	SAT Super Score & Single Score: 1580 & 1560

TECHNICAL SKILLS

Languages: Proficient in Python, JavaScript, SQL, HTML, CSS, and Java
Frameworks/Libraries: React, FastAPI, Vite, LangChain, Tailwind CSS, Node.js, PyTorch, and Pandas
AI/Machine Learning: LLM Integration, Prompt Engineering, RAG (Currently building), Reinforcement Learning (PPO, DQN), CNNs, LSTMs
Databases & Services: PostgreSQL, Supabase (Auth, RLS, DB Functions), Pinecone (Currently building), OpenAI API, Vercel, Git, GitHub
Videography/Photography/Editing: Davinci Resolve, Krita, Adobe Creative Suite, Canva
Manufacturing: MIG Welding, Angle Grinding, 3D Printing, Manual Mill Machining
3D Design: Fusion 360 CAD, Onshape, SolidWorks, Ultimaker Cura, Afinia 3D

PROJECTS & EXPERIENCE

Recall AI, a Full-Stack AI Study Platform July – August 2025
Visit the Vercel page here — Demo login is {Email: "randomlexicon81@gmail.com", Password: "pass123"}

- Architected and deployed a full-stack application using FastAPI's serverless API, hosted on Vercel
- Engineered a context-aware AI tutoring chatbot using OpenAI's GPT-4o, featuring persistent conversation history via localStorage and client-side logic that minimizes token consumption on subsequent messages using message summaries
- Developed a generation pipeline using LangChain to enforce structured JSON outputs from the LLM, ensuring data integrity and eliminating the need for fragile string parsing
- Implemented a robust "cache-then-database" data synchronization strategy, prioritizing fast loads from local cache while ensuring data consistency by falling back to a PostgreSQL database as the single source of truth
- Secured the multi-user database at the data layer by configuring PostgreSQL Row Level Security (RLS) policies, guaranteeing users can only access and modify their own data
- Engineered a secure, multi-step deck deletion process with a confirmation modal, handled by a dedicated backend endpoint that atomically removes data after verifying user ownership

Autonomous Driving Research Intern, University of Texas at Dallas June – September 2024

Student Intern & First Author

- First Authored a machine learning research paper and published in the Curieux Academic Journal
- Worked with remote servers, cloud-based data, and the TensorFlow library to build and train ML models
- Led a team of 6 students to research and develop a state-of-the-art object-detection ML model to accurately locate vehicles, pedestrians, and other obstructions in real time

National Solar Car Challenge August 2022 – Present

Co-Founder & Co-Captain of the team

- Founded the Coppell Solar Car team of 9 and hosted meetings to plan, fundraise, and build the car
- Built a fully functional (steering, electric motor, brakes, suspension, electrical system, etc.) 1-seater solar-powered electric vehicle from scratch (top speed of 45mph)
- Registered the team as a 501(c)(3) non-profit, raising over $6,500 in parts and funds through local & national company sponsorships
- Competed in the National Solar Car Challenge race at the Texas Motor Speedway in 2025 and 2024, placing 4th nationally as a second-year team

Captain of the Design Team

- Designed the chassis, crush zone, and panel array supports
- Co-designed chain-drive components and rear-wheel supports
- Led CAD workshops to teach the team using Fusion 360

Python Coding Projects ~ (visit pranavgupta.co to see more) June 2021 – Present

- Developed over 100 custom projects in Python, focusing on AI simulations, physics engines, procedural generation, and video games using Pygame as a visual interface.
- Built and trained a PPO-based reinforcement learning agent with PyTorch to control a drone in a custom 2D physics simulation
- Created a cloth physics, particle physics, and a particle-based fluid dynamics simulation
- Built dozens of top-down, platformer, two-player, RTS, and turn-based games using a variety if input streams (hand gestures through a video camera, voice-activated movement, and KB&M)
- Built visual-effects renderers including OpenGL integration for 2D shaders, a Delaunay Triangulation-based mesh generator, and ray-casting with reflections

Boys Scouts (BSA, Troop 840), Coppell, TX September 2017 – Present

Eagle Scout at Troop 840

- Completed 24 merit badges, 14 of which are Eagle Required, including first aid, archery, robotics, and environmental science Den Chief at Pack 841

Rocketry Club SEM High School September 2023 – Present

Co-Founder and Captain of Rocketry Club

- Organized after-school lessons to design, model, build, and launch over a dozen model rockets
- Taught members about aerodynamic and physics principles

AWARDS & ACHIEVEMENTS

Awards & Honors: National Merit Scholarship Finalist; 3x UIL State Champion (Awards in Math & Science).
Robotics (FRC & VEX): Contributed to mechanical/software development for a FIRST Robotics world-championship competing team; led VEX Robotics team to a 2nd place regional victory as Co-Vice President.
Eagle Scout (BSA): Managed a yearlong community project to fundraise $1,500+ for, CAD model, and construct new playground facilities for a local elementary school
Academic Scores: SAT: 1580; Completed 16 AP Exams, achieving 14 5/5s in subjects including Calculus BC, Physics C (E&M, Mechanics), and Computer Science A